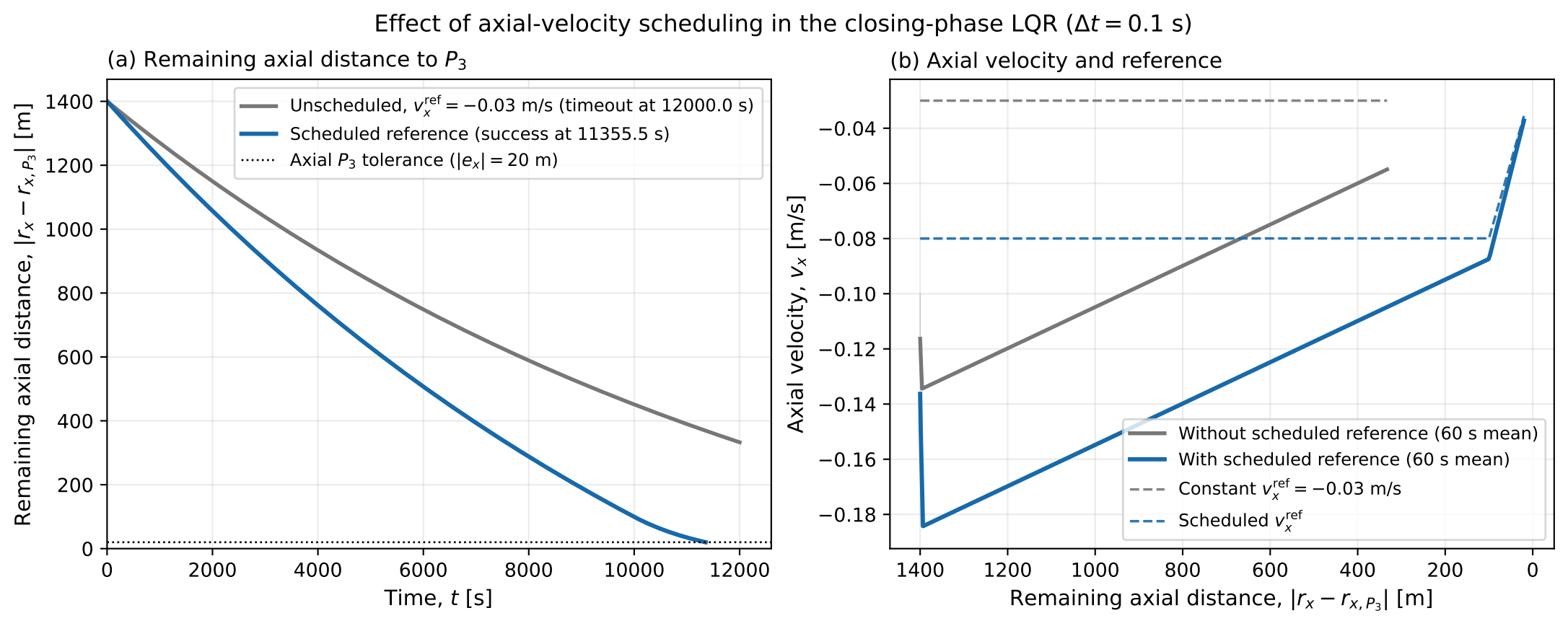
Effect of axial-velocity scheduling in the closing-phase LQR ($\Delta t = 0.1$ s)
(a) Remaining axial distance to P_3
Unscheduled, $v_x^{\mathrm{ref}} = -0.03$ m/s (timeout at 12000.0 s)
Scheduled reference (success at 11355.5 s)
Axial P_3 tolerance ($|e_x| = 20$ m)
Remaining axial distance, $|r_x - r_{x,P_3}|$ [m]
Time, t [s]
0
200
400
600
800
1000
1200
1400
0
2000
4000
6000
8000
10000
12000
(b) Axial velocity and reference
Without scheduled reference (60 s mean)
With scheduled reference (60 s mean)
Constant $v_x^{\mathrm{ref}} = -0.03$ m/s
Scheduled v_x^{ref}
Axial velocity, v_x [m/s]
−0.04
−0.06
−0.08
−0.10
−0.12
−0.14
−0.16
−0.18
1400
1200
1000
800
600
400
200
0
Remaining axial distance, $|r_x - r_{x,P_3}|$ [m]